

A Theory of Individuality

Hansjörg von Gemmingen has struck a world record for the longest driven Tesla ever for over 2 million kilometers in 2024. To score such record, he has changed the electric motor 13 times, the battery 3 times. For this reason, it is a well-known joke in the community to ask if it is still the same car that Hansjörg bought in 2014 (Singh). This funny joke ponders no more peculiar question than to ask whether a person today is the same as the person yesterday, or the Theseus's ship reconstructed in the warehouse is the same as the original. Both questions pinpoint only one concern: Are different things by nature, *different*?

For people who are not familiar with Theseus's paradox, the Ship of Theseus is a famous thought experiment in philosophy that questions the nature of identity over time. According to legend, the Athenians preserved the ship on which the hero Theseus once sailed. As the years passed, its wooden planks began to rot, and one by one, they were replaced with new timber **gradually** over years until eventually no original piece of the ship remained. This replacement raised a puzzle: was the restored vessel, built entirely of new parts, still the same ship of Theseus? The paradox deepens when philosophers imagine a scenario where someone gathered all the discarded, original planks and reassembled them into a second ship. At this point, there appear to be two ships - one made of the original materials but reconstructed later, and one that has existed continuously but with entirely new parts. The question then becomes: which of the two, if either, is the true Ship of Theseus (Carroll and Markosian)?

Note that when I say true ship of Theseus, to the best of my understanding, the term **identical** is *ambiguous*, as there appear to be at least two types of identities the word represents: narrative identity, and metaphysical identity. Narrative identity is the daily concept that people use in language to register something *identical* in functions; and the metaphysical identity is

about whether two things are **actually** identical. In the daily convention, it is sometimes enough to say two things are identical if they perform similar functions. For example, if I say, “This is the same bus that I took yesterday from Knoxville to Sevierville, I might not mean the **actual** same bus but rather a different colored bus runs the similar route with different bus driver, passengers, etc.” On the other hand, in metaphysical identity notion, if I say “This is the same bus that I took yesterday,” what I mean is the bus is **literally** the same bus with the same drivers, same passengers, etc. For the sake of clarity, it is necessary to specify which type of identity being discussed at the moment. If not specified, audience can understand in either way. With the now disclaimed identity concept, we dive deep into Theseus’s paradox.

The Either Argument

Philosophical discussions of the Ship of Theseus have produced several possible answers to its central question. Considering the following argument: (i) the ship docked, referred to as Ship A, is the true Ship of Theseus, or the ship reconstructed in the warehouse, Ship B, is the original - an approach I call the Either argument. In this argument, it is assumed that individuality is unique; and that the ship of Theseus can only be one of the two ships. Although this argument aligns with the traditional view upon individuality about uniqueness, the approach is *vague* because it has NOT established well the criteria used to distinguish between the two: if one believes the docked ship to be the original ship, he is as well reasonable to say the same for the ship at the warehouse; both ships are original in a sense and there is simply no measurement to prioritize one over another. On one hand, the docked ship *has been* the Theseus’s ship while the ship at the warehouse composes of all the original pieces *at the end*. If we believe ship A to be the origin, then what do we say about ship B with **all** the original pieces and ship A with **none** of the original piece? If we believe ship B to be the origin, then at what point does identity

transfer from ship A to ship B? Does identity transfer in continuous pattern, or does it jump? If identity is discontinuous, then what is the *space* between the transformation? These questions are not easy to answer, which makes the Either argument vulnerable in many aspects. To make it easier to answer the central question, how about we reconstruct the Theseus's story in another way. Instead of one ship with zero original plank and one ship with complete original planks, Athenians build a warehouse ship with exactly **fifty percent** of *arbitrary unit*, which could be the area, the number of planks, etc., of the docked ship, how would the Either argument respond to this example?

At the ground level, this modified version of the old story introduces more complicated considerations, shaking the binary nature of the Either argument. This binary belief comes from the assumption that the individuality is unique so that no two things can be one at any moment (Leighton 567). Therefore, if we were to equate identity of objects to uniqueness, then we would have problems favoring one ship over another when both are exact half of the original. One can circumvent this problem by claiming a specific value that defines "originality", i.e. 98% versus 100%. If 2% is the significant difference that changes the ship identity, then might we be able to dismiss the paradox? I see that at least two problems arise when we employ such connotation for identity. Firstly, if they're *slightly* different, but not enough to make *significant* difference so we equate two things one, it is then an example of *narrative identity*. In this type of identity, two things need not be totally the same, just functionally the same, meaning they can lack a thing or two but still announced identical. However, as mentioned in the beginning, this paper concerns more on the *metaphysical identity* where we reflect whether two things are **literally** identical. Second problem arises due to the mathematical understanding of transition. Let's consider three cases where ships A has 100%, 99%, and 98% of the original Theseus's ship, correspondingly to

case 1, 2, and 3. Assuming that 2% is what makes the *metaphysical difference* between two things, then case 1 is identical to case 2 and case 2 is identical to case 3; but case 1 is *significantly* different from case 3. In traditional comprehension of mathematics, it is the *transitive property of equality* warrants that case 1 and case 2 must be also *significantly* different for case 1 and case 3 to be *significantly* different in this example. By similar logic, one can see that regardless of any milestone claimed to be *significant*, there are always cases to dismiss such statement. In fact, this is one of the so many called examples of **Sorites paradox** (Clark 86).

The Sorites paradox, or heap paradox, is a classic philosophical problem about vagueness that arises when small changes occur without a clear point of transition. It is often illustrated using the example of a heap of sand: we accept that 10,000 grains of sand arranged together form a heap, and we also agree that removing just one grain does not stop it from being a heap. However, if we repeat this reasoning grain by grain, we end up concluding – falsely - that even a single grain of sand is a heap. This problem is not limited to heaps; it applies to many vague concepts such as “bald, tall, adult, mountain, etc., and in light of our paper: “identity.” All these words lack precise boundaries for their application, so they more than often fail to yield any definitive answer to tackle on the crisis, making the paradox incredibly tough against attacks. Nevertheless, we cannot discuss identity without attempting to batter Sorites paradox at first.

There have been multiple proposals from philosophers to dismiss the paradox, each comes with distinct sets of advantages and disadvantages. *Ceteris paribus*, let us now examine some of these views upon the problem.

Epistemic View

The problematic nature of Sorites paradox can be illustrated in logical form, starting with a specific case of a bald person, adapted from Ofra Magidor's paper.

- (i) A person with no hair filament is bald
- (ii) For n hair filament ($n=0$) is bald, every person with $n_i = n+1$ hair filament is also bald

Therefore,

C: A person with n_i hair filament must be bald as well.

Nevertheless, as we know, a person with thick, healthy hair is not "bald" at all. Because the conclusion is false, we assume that one of the two premises must be false. In epistemic view, (i) is *undeniably* true, as we all agree that zero hair *absolutely* means "bald." Therefore, it only makes sense to deny on the second premise. According to epistemicists, there is a sharp cut-off point in the example, it's just that we do not and cannot *know* precisely where it is.

A great advantage of this explanation towards the paradox must be the fact that they align quite well with classical logics and semantics, meaning they obey the binary nature of true or false, bald or not bald. People can either be bald or not bald, and these kinds of statements can only be true or not true regarding to our state of *knowing*. For this reason, I want to attempt on employing epistemology to discuss the Sorites paradox, which according to me is a convenient method to understand little-by-little argument.

How do we tell something exists? To say something exists is to at least somewhat recognize the uniqueness of it over others. I will employ the scientific terminology here and say that the ability to tell things apart from others is called *resolution*. For example, our eyes do not possess great resolution to tell the difference between two so small particles, so we need an

intervention from an optical instrument to see the micro world. In analogy, I think that our minds have too some sort of *resolution* and in desperate need of an instrument helping us to see. But what is it we need to tell the difference apart?

Before I attempt to answer the first question, I need to point out that metaphysical identity is a metaphysical concept, and only metaphysical means can meet their ends. Hence, I seek NOT define identity with the traits only available to humans or animals such as morality, psychology, or animalism. Although I believe these theories cover a wide range of specificity, they do not generalize enough to stand out as the universal principle. Take animalism for example (Bailey), while I acknowledge that his argument is appealing, animalism cannot explain the identity of trees, inorganic objects such as stones, or ships. For that reason, I will not even bother attempting animalism to explain identities of inorganic objects, let alone identity of *everything*. Let us start the work with Hume, a Scottish philosopher whose work in “Of the Idea of Necessary Connexion” establishes a solid ground for examining causation. According to Hume, causation does not have an objective root in which causality happens externally to the minds. Instead, causality is built upon repetition of habits and customs of the observers. In his own words, Hume enunciates:

a necessary connexion obtaining between the present fact and that which is
inferred from it as cause or effect. Were there no causal connection between
the observed and the unobserved, were there nothing to bind them together,
the inference would be entirely precarious and unwarranted. **Were watches
not regarded as creations of watchmakers, there would be no reason to infer**

the existence of a watchmaker from the existence of a watch. (Stapleford & Goldschmidt 47)

In Humeian logic, there is no objective causation to the watches besides the *constant conjunctions* in which people perceive all past instances of the watches in association with the watchmaker. It is these experience and observation together define the causal relation between the watchmaker and the watches in which one effect happens after another in a sequel. In other words, the very existence of the watches and the watchmaker depends on the ability of the minds to *differentiate* between two effects. If the conjunction is not clear, if the watchmaker had not been clearly the one present in every instance of the watches's creation, our minds would have been twisted and confused the two effects as **narratively identical**. The mind constructs narrative identity in this way because it requires a clear causal thread to distinguish between phenomena: to tell a tiger from a deer, or a bear from a squirrel, which has been essential for survival throughout human history. In narrative identity, *the minds cannot see the differences between two effects unless they can register the primary cause*, a pattern Anscombe called deterministic in the human minds (Anscombe & Ekstrom 59). While narrative identity has served as a powerful tool for human survival, it simultaneously narrows our comprehension of higher understandings. Take history for example, there are multiple causes leading to the fall of Roman empires, including military unresponsive actions (Ferrill), loss of liberty (Piganiol), or environmental factors (Harper) but I guarantee that none of these is the *actual* cause for the Roman empire loss. Historians must study and *determine* a cause in their minds to understand and *differentiate* the fall of Roman empire from the fall of similar empires such as Persian empires, Ottoman empires, or the British empires. This trend is not unique to history study, since

if we construct cause-and-effect in a Sorites, we expect to see a *universal* trend of having no clear point of transformation. Take eating a sandwich for instance, where:

- (1) I am eating a sandwich, which is an effect in consideration.
- (2) This effect is caused by a person cooking and prepping the sandwich for me
- (3) The cooking and prepping can only happen because a truck delivers ingredients to the restaurant
- (4) The delivery is also in turned caused by gasoline provider that makes the truck able to run
- (5) And so on...

If we construct the causal relationship in a Sorites, it will seem that there is no clear point to stop, no cause is the *real cause* to an effect. *Without the minds registering the primary cause, all effects are infinitely caused*; and consequently, effect 1 is not well-differentiated from effect 2, which is responsible for infamous Sorites paradox. In other words, as opposed to what epistemicists proposed, there is a clear point in the transformation between two effects in epistemological approach, in which narrative identity heavily depends on the active power of the minds and how well it acknowledges the *primary cause*.

Although the causal chain appears infinite when considered in abstract terms, the mind intervenes as an active agent that determines where the sequence “tips” into meaning. In this sense, the mind functions as a **cognitive tipping point** in the construction of causality. It decides when and where to terminate the endless regress of causes, transforming what would otherwise be an indeterminate sequence of external events into a coherent narrative of cause and effect.

In this way, the mind does not passively record external reality but actively shapes it, disrupting the infinite regress of causes by “tipping” the system into an interpretable framework. External objects and events, in themselves endlessly entangled, only become intelligible when filtered through the mind’s selective acknowledgment of a *primary cause*. Thus, narrative identity and causation are not merely features of the external world but emerge as internal reflections structured by the observer. This tipping point in the system highlights that human cognition introduces critical thresholds where continuity is broken into discrete narratives, and where **disorder** is transformed into **order**.

Since *primary cause* itself is a type of knowledge, it does not avoid or lessen the problems of epistemology on whether we can obtain knowledge. A lot of epistemologists have debated this possibility for a long time (Chisholm, *The Problem of Criterion*) while the others disagree even on the definition of knowledge: Gettier examples versus the Traditional Analysis of Knowledge (Feldman). Gettier examples are special cases of traditional analysis of knowledge where only by chance a **justified true belief** is true, which counters the traditional definition about knowledge and classify it as *insufficient*. One significant thing to note about Gettier’s cases is that they are inescapable (Zagzebski), thus render the answer to the question whether we can obtain knowledge even more complexed, probably luck-dependent too. When we don’t know for sure the possibility of knowledge, or more specifically, the *primary cause*, it’s controversial to say anything about the narrative identity of objects, as it seems too much *arbitrary* to say one thing is the same or different from another. Due to close association between epistemology and narrative identity, any reply to the Sorites paradox can also respond to the theme question in epistemology: “Can we have knowledge?” and vice versa. In short, although the epistemological view provides a definite solution to the Sorites paradox, the view itself does not escape the

bigger problems inherited from epistemology. For this reason, I think it would be inadequate to batter Sorites from the knowledge viewpoint alone, thus I am also attempting to counter the Sorites paradox by introducing the “Degrees of Truth” concept (Clark, The Heap, 89).

Degrees of Truth

Degrees of Truth argues that anything on the spectrum is precisely the probability of truth with the extreme cases probability most ideally start with 1 and shift towards zero at the end of the spectrum. For example, a no hair person would be 100% chance to be bald, with less chance of bald as the statements shifts further away from the non-hair pole until he reaches 0% chance of being bald at the infinite-hair pole. Qualitatively, degrees of truth argument have its peak at solving the cut-off problem since it claims that there is no cut-off point but only a directed scale of probability throughout the gradual shift. In this sense, instead of a binary pattern, a person with n hair filament only has more chance to be bald than the person with $n + 1$ hair filament but *need not* to be bald. This probability argument appears to be *sound* to some extent. However, this argument soon tears apart when one tries to assign a numerical value for the likelihood of positions on the spectrum. Let's consider Kent is a man with absolutely no hair filament with a likelihood of being bald is 1, then what is the numerical likelihood of being bald of Norman, who is next in line with 1 hair filament? The most ideal solution would be greater than .5 and less than 1 (Clark 90) but anything falls in between this range is quite artificial. If one says that Norman has .8 chance of being bald, then it is not clear on *why* it must be .8 instead of .99 or .98. This illustration is known as the higher-order vagueness problem: **we do not have a clear cut-off point for the probability of truth.**

Now, imagine Norman with one single hair filament walking down the street and talking with 100 people, of which there is one person Norman successfully convinced that he is NOT

bald as he still has one hair filament left. To Norman, he has a 99% chance of being bald; and to the convinced person, Norman has 0% chance of being bald. The interesting point is here: to the convinced person, any person with more than 1 hair filament from now on will not be considered *bald* thanks to Norman's influence. In other words, to the convinced person, there is a clear cut-off point between *bald* and *not bald* by having *one* more hair filament; and his spectrum contains only two extremes of baldness with zero hair filament at a 100% bald and one hair filament with a zero chance of bald. According to Clark (92), this establishment happens because **borderline cases are those which we hesitate to classify, but the point at which we start to hesitate will typically vary from time to time and person to person**. To put it another way, the higher order of vagueness can be alleviated by employing the powers of the minds. *Once more, one can circumvent the Sorites paradox by recognizing the different boundaries in the minds of people.*

There is a strong connection between the minds and the answer to the Sorites paradox, which in turn heavily depends on the knowledge, which sometimes can be obtained in the luck fashion (Gettier examples). The inescapability of Gettier's problems in knowledge-seeking (Zagzebski) suggests that solutions to the Sorites paradox can sometimes come by luck; and since narrative identity over time can be constructed in a Sorites, which entails that distinguishing between narrative identities requires *knowledge*. I want to talk about the metaphysical identity in this paper, but we cannot discuss how the pyramid is built without paying attention to the building bricks. The same relation happens here: although metaphysical identity is the external identity of things, but people cannot see the whole, only a manifestation of it in the mind. We can't talk about metaphysical identity without clarifying on the narrative identity at first.

It is now a good time to revisit the central question posed earlier concerning the continuity or discontinuity of identity. If identity is a continuous spectrum, like any spectrum exists continuously with *quantized* amounts (in discrete number) of fundamental units of that spectrum, identity itself also has a fundamental unit that needs a cautious look. Let's consider baldness for instance, the position on the spectrum can be defined with respect to how many hair filaments the person has; color spectrum that can be defined with respect to the wavelength of electromagnetic waves; or energy spectrum is in *quantized* number of elementary particles such as electrons, photons, gravitons, etc., which emit waves displayed themselves as energy fields.

If, on the contrary, identity is discontinuous, then we must account for what occupies the intervals between successive identity states. To study this problem and seek a more adequate answer, might it be instructive to examine a classical paradox that has long challenged metaphysical accounts of identity constitution.

In *Paradoxes from A to Z*, the author mentions "The Paradox of the Many" (Clark 131) where Tibbles the poor cat seems to have too many identities and thus experience an existential crisis. The paradox goes like this:

Tibbles the cat is sitting on the mat. Tibbles minus any one of 1,000 hairs is also a cat. If any one of those hairs had been missing, we should still have had a cat there on the mat.

So, it seems we have 1,001 different cats on the mat, for Tibbles has precisely the hairs he does at the time and so cannot be identical with any of the cats with only 999 hairs.

(Clark 131)

The paradox arises for which it suggests Tibbles is not a single cat but a collection of countless slightly different cats. Yet it is clear that Tibbles is an individual, not a feline transformer made of

a thousand cats, each with one hair less than the previous. Thanks to this counter-intuitive speculation, Tibbles the philosopher cat is crucial to determine the property of identity.

Assume identity is discontinuous, it entails that there is at least an instance Tibbles' identity is lost and recovered. For elaboration, is there any possible scenario that Tibbles with 1,000 hairs is Tibbles, not Tibbles with 999 hairs but Tibbles again with 998 hairs? If there was such an instance, we would have no other choice but to conclude that identity is discontinuous; otherwise, we must say that identity is a continuous spectrum.

To analyze the scenario of Tibbles and disprove the notion of discontinuous identity, one must emphasize the implausibility of identity being lost and then later "recovered." If Tibbles with 1,000 hairs is truly Tibbles, but Tibbles with 999 hairs is not, only to regain his identity at 998 hairs, then identity would rest on an arbitrary cut-off point. Such a position would be incoherent, for it would require that at the precise moment one hair is lost, Tibbles ceases to exist, only to spontaneously reappear after. Nothing in our ordinary experience of objects and organisms justifies this bizarre oscillation of being and non-being, exist and non-exist. Therefore, by *reductio ad absurdum*, the more coherent interpretation is that **identity is continuous**.

Tibbles' identity does not vanish with the loss of one hair, nor does it suddenly re-emerge with the loss of another. Rather, identity persists through gradual change, even if the boundaries of what counts as "Tibbles" are somewhat vague. This continuity reflects the ordinary way we understand persistence: humans remain the same individuals despite the constant turnover of cells, and ships remain the same vessels even when their planks are replaced. Thus, Tibbles at 1,000 hairs, 999 hairs, or 998 hairs remains Tibbles, for identity is not defined by a discontinuous threshold but by the endurance of the object across incremental variation. The conclusion,

therefore, is that identity is best understood as a continuous spectrum whose transformation occurs in the minds of observers, not as a series of abrupt disjunctions.

However, “ordinary” understanding about identity does not subdue the “many identities” challenge. For even if Tibbles remains Tibbles with each successive hair loss, it implies that his identity is preserved across a series of overlapping candidates: Tibbles with 1,000 hairs, Tibbles with 999 hairs, Tibbles with 998 hairs, and so on. But on this account, the *loss* of a single hair *gains* the original Tibbles with 1,000 hairs new identity as Tibbles the cat with 999 hairs. While true, this peculiar logic creates the peculiar result that, while we confidently say there is one cat on the mat, our logic appears to commit us to acknowledging a multitude of nearly indistinguishable cats, each with a slightly different constitution. The problem, then, lies not in the question about continuity of identity itself but in reconciling this continuity with the multiplicity generated by ordinary reasoning.

As part of counter argument to the “ordinary” reasoning, one could try to avoid the puzzle by stipulating that Tibbles is defined precisely by 1,000 hairs. However, this definition fails as soon as he loses a hair, for he remains recognizably the same cat. Hence, Tibbles’s identity cannot rest on an exact number of hairs but must instead be allowed as a range, suggesting that Tibbles is a superfluid existence (Clark 131). This realization forces us to consider identity in terms of fluid rather than solid where identity becomes less a static property and more a dynamic continuity that accommodates variations while preserving recognition for individuals. Knowing *narrative identity* is continuous, fluid in nature, and strongly associated with knowledge, what could its fundamental unit be?

In my view, this fundamental unit must be **information**.

If information were to be considered a fundamental unit for identity of objects, it would be the *necessity* of this world because the identity of objects founds the root on which the world is built. But is there any *necessary* factor in any world to begin with?

It is helpful at this point to bring in van Inwagen, whose work in “Why Is There Anything at All” discusses quite detailed the existence of *something* in the world. In the paper, Inwagen agrees that “the probability of there being no beings is 0” (Inwagen 100), meaning there must be a *necessary* factor constructing this world. I will name this view about *necessity* **fundamentality**, meaning there is a fundamental law setting up the universe as we know. In common sense, there are two modes of fundamentality: (α) fundamentality is caused, then it entails that fundamentality is infinitely caused by a chain of hierarchical fundamental principles, resulting in a regression of fundamental laws in which we shall find no end to the series; or (β) we must terminate the chain in exchange for one final uncaused fundamentality. Say we accept the former, we then inevitably admit there is no trace back to the origin of any world, unclear on *how and why* things exist. If he were to be alive, Nietzsche would be very interested to voice in a lecture on how problematic nihilism is and how to overcome it. On the other hand, if we accept the latter, we must accept an uncaused fundamentality, a God above all, and we circulate right back to the traditional arguments about existence of an omnipotent, omniscient, and wholly good God.

Sui generis, I think there is (γ) a third mode of understanding upon fundamentality, one that is contradictory by nature and cycled through caused and uncaused sequence. For elaboration, a universal God’s existence (an uncaused fundamentality) is commonly denied on the ground of NOT omnipotent, or omniscient, or wholly good because these holy trinity are quite often conflicting with each other and cannot fulfill the requirements for The Most High

Being. The most typical attack is The Problem of Evil: if God is truly omnipotent and wholly good, why does he allow for evil and suffering to exist? No matter what the responses from the theists are, might it be *the good of the apparent evils* or *the lesser of the two evils*, they all demonstrate to some extent a flaw in God's ability to unite one explanation to human beings, almost paradoxical to human's knowledge. Some people say paradoxes about God happen because humans are not omniscient, thus the paradoxes so far only illustrate the limitation in mortal knowledge about The Most High Being. Specifically, philosophers say this paradox originates from deep flaws either in logics or semantics rooted from the language usage (Schrader and Savage). What if the issues are not the language of humans, but rather a nature of whatever ultimate **fundamentality** that guides this world is indeed *self-conflicting*? For example, is a universal God powerful enough to create a stone that even he cannot lift it? Atheists often use this example to deny the existence of an uncaused high being, an omnipotent God on the ground that he cannot create such stone, for which the action violates the definition of being all-powerful. I want to be creative here and say that an almighty God with uncaused existence can create such blasphemous stone, but in doing so, the stone causes denial to his existence so that God's existence is both caused and uncaused simultaneously. I am not sure if *self-conflicting* is the right word to use here because so far as I understand, an omnipotent being is also omnipresent, meaning he can be anybody all at once. If we understand the stone paradox in a way, such that:

- (1) Omnipotent God is composed of Stonecraftius and Rockules
- (2) Stonecraftius can create any stone
- (3) Rockules can lift any stone

Because these two people have *infinite* amount of time, if Stonecraftius can create a stone that Rockules is not able to lift, Rockules will immediately lock in and train until he can lift the stone and vice versa. Voilà, the whole paradox just turned into a recursive competition between Stonecraftius and Rockules, who are constantly working to win each other! Suddenly, impossible becomes possible, paradoxical becomes logical, and *one becomes many*. In conclusion, should there be an ultimate fundamentality, it must be *self-contradictory*.

Information too has this critical characteristic. This point is exemplified by the historical episode in which Caliph Uthman authorized the compilation of an official recension of the Qur'an while ordering the destruction of all other textual variants (Olatunde). Such actions appear to constitute a straightforward case of loss, in which significant manuscripts were removed from the record. Yet the irony resides in the fact that the very act of destruction conveys *information*, preserving the trace that alternative versions of the text once existed. In this sense, **destruction** functions as **creation**: by extinguishing textual diversity, Uthman inscribed the materials of that diversity into historical memory. This example illustrates how information experiences *self-conflicting*.

A similar self-conflicting dynamic emerges in a more mundane but equally illustrative scenario: imagine a philosopher who spends eight hours in a day working diligently on an essay he has been composing for months. By the end of the day, he has written three new paragraphs but has also removed four that he now deems irrelevant. The question naturally arises: has he made progress, or has he regressed? On one hand, the essay has become shorter, objectively diminished in length. Yet on the other, its conceptual clarity and intellectual precision may have improved. The act of deletion, which appears as a loss, simultaneously functions as a form of addition, refining the text into a more coherent whole. Thus, the philosopher's work embodies

the same paradox seen in information itself: creation through destruction, addition through subtraction, and progress through apparent retreat. In conclusion, although self-contradiction may be regarded as a necessary feature of any ultimate fundamentality, it alone cannot secure explanatory adequacy. What is still required is a *sufficient* condition, something that does not merely posit existence in contradiction, but also accounts for how such a fundamentality can sustain and generate the reality we inhabit.

My *sufficient* comprehension about information comes from a time I trample upon a thought-provoking work authored by Dr. Yuval Noah Harari named *Nexus: A Brief History of Information Networks from the Stone Age to AI*. As a summary of chapter 1, the author criticizes upon a “naïve view” of **information** - the belief that information exists solely to depict or mirror objective reality. In other words, it would be naïve if one sees gray clouds coming in and tells his friends “gray clouds coming in,” thinking that would suffice to mirror an objective reality. Need not mentioning true or false, a piece of information, no matter how detailed and honest it is, always omits other crucial parts of reality. For example, if I see a couple eating in a restaurant and I tell my friends “There is a couple eating next to us.” The sentence is a complete piece of information as my friend now knows there is a couple eating next to him. Yet, in delivering this sentence, I have inevitably excluded a multitude of details that are equally real and significant: their physical appearance, the nature of the food they are consuming, their body language, the duration of their meal, the emotional atmosphere between them, or even the cultural context of the dining setting, etc.

It would even be more naïve to think that the conducting sentence needs to be true to be classified as information. What if I am mistaken? Suppose I was mistaken in my assumption, and the individuals I identified as a romantic couple are, in fact, siblings sharing a meal? Assume that

my friends did not know anything about two people I mention, the only plausible option they have is to believe my words until they receive contradictory evidence. In such cases, we typically describe the information as misleading; yet the crucial point remains that the statement still functions as a source of information. Assume my friends are blind so my voice is the sole source of information my friends can access, it then becomes undeniably true for them that two people in question dining together constitute a “romantic couple.” From what I see, the traditional notion that information must be factually true to be meaningful or consequential is inadequate. Instead, a single piece of information, in the absence of competing sources, even if inaccurate, can dominate and define an individual’s understanding of a given reality.

This concept extends beyond human communication into the biological realm, where the language of life itself - DNA - the fundamental source of living information, also witnesses this phenomenon. In molecular biology, DNA sequences encode genetic information that directs the structure and function of living organisms, known as central dogma of modern biology. Importantly, mutations - random alterations in the DNA sequence - are sources of novel genetic information. These changes give rise to unprecedented biological traits, thereby serving as the raw material for evolutionary processes. For instance, the emergence of antibiotic resistance in bacteria is the result of genetic mutations that confer survival advantages in antibiotic environments. Mutations, the random changes in genetic information, divide the original organism into two different individuals after the changing point: one that resists antibiotics, and one that does not. Essentially, alterations in genetic information are not mere reflections of an already existing example elsewhere, but acts of creation, producing novel forms of life. In this way, what the world had never encountered before now comes into existence and attains the *status quo*.

Natural evolution then may be conceived as the process by which individuals bearing a particular trait, such as an antibiotic resistance gene, are preserved while those lacking it are eliminated from the genetic pool. In this sense, the DNA of unfit organisms is effaced, and the continuity of life is carried forward only by those whose genetic constitution confers survival. Evolution is a sequence of many actions act as sources of information, a multiplicity of informational streams layered upon a prior state of reality (mutation), reshaping it and establishing a new order of truth, later became what we know as history. Within this framework, the transition from susceptibility to resistance exemplifies a transformation in narrative identity: a bacterial strain that once lacked, for instance, the Kanamycin resistance gene (KanR) may, through mutation and subsequent selection, acquires it. In so doing, the identity of the organism shifts, not merely in a biological registry but also on the identity spectrum, as its very being is redefined by the addition or reduction of new or existing informational content.

The multiplicity of informational streams layered upon reality can be understood as what van Inwagen describes as **maximal specific states of Reality** (Inwagen 102): complete ways the world might be, each internally coherent and self-contained. Every informational shift may be seen as a state overwriting on another, reshaping what counts as Reality to our perception. In this sense, informational change does not merely alter the present world but represents the potential actualization of possible worlds, suggesting that Reality is not a singular fixed construction but rather *a field of informationally possible configurations superimposed on each other*.

For this reason, Reality often discloses itself not as a single, fixed entity but as a convergence of perceptions, each rooted in its own informational frame. Two people may inhabit the same space, share the same air, and yet perceive entirely different worlds. Imagine, for instance, a dimly lit room where shadows shift and movements blur at the edge of vision. You

glance beneath the table and catch sight of what seems to be a roach darting into the darkness.

Your brother, standing across from you, looks and sees nothing. He insists the floor is empty, that your mind has played a trick on you. In that moment, two versions of reality coexist: one in which the roach exists, and another in which it does not. Each is internally consistent, sustained by its own source of information: your eyes versus his. The objective Reality, then, is the composite of informational multiplicities that unite to form what we call “the world.” **In conclusion, information is not true or false, nor does it depict or mirror partial, or total reality. It is Reality (Harari).**

Since information is materials for Reality, and narrative identity of things is the building blocks of this Reality, hence *information* is the core for *the principle of identity formation*.

If identity is to be understood in terms of information, then two corollaries follow. First, the narrative identity of an object may be defined by the quantity of information it contains or is perceived to contain. Second, since no two objects embody precisely the same informational content, individuality arises from the varying degrees of informational density each object holds. On this account, identity may be conceptualized as the aggregate of informational content, which I designate as the ibit (identity bit). The distinction between one object and another, then, is captured in the differential Δ ibit, representing the informational gap. Should this framework hold, it entails that every identifiable entity necessarily occupies a distinct position on the continuum of Reality.

This formulation yields two faces of a coin. The head face is that, insofar as identity is the total sum of information, **it is possible for one entity to subsume or overlap with the identity of another**. This overlapping is what happens with Tibbles the philosophical cat, where Tibbles with 1,000 hairs overlap his identity with “other cats” with less hairs, forming a superfluid

existence named Tibbles, forming what we refer to as “Paradox of The Many.” This trend does not just stop in theory, but rather extend to a paradigmatic observation in science, where the relation between humans and chimpanzees: genetically, humans differ by only 1.6 percent of DNA (Ebersberger et. al., 2002), and thus we are said to be 98.4 percent “identical.” I employ “identical” here in the technical sense used by molecular biologists to denote the degree of correspondence between genetic contents. Yet, genetic information is only one species of information; beyond it, there exists epigenetic, cultural, contextual, and other layers of information, all of which contribute to the constitution of identity. Taken together, these strata imply that the identity of one entity may partially contain, and thereby be continuous with, the identity of another.

The tail face of this formulation says that *nothing* is ever *narratively identical*. The part that one identity does not coincide relatively with another is the *individuality* of that object, thus gives the unique nature of one over another. I think there is no one better than duo Dion & Theon to explore this question. Take the author’s words for illustration:

Theon consists of Dion except for Dion’s left foot. If Dion’s left foot is amputated, how many men do we have? Two? Dion can survive the amputation and Theon has been unaffected. Has Theon been turned from part of a man into a whole man, identical but distinct from Dion? (Clark 132)

From the point of the question, it seems like we have an interesting case, where Theon now is no different from amputated Dion, and two individuals *become one*. To circumvent this problem, we can “recognize... that ‘Theon’ was never the name of a man, but of a physical part of a man, which, together with left foot, made up the body of Dion” (Clark 132). Here, I notice how the solution to the problem yields an unconventional grasp upon identity: Theon, if he was a man, he would be a part of another man named Dion. The difference between these two men narrows down to Dion’s

left foot for what defines Theon's unique nature is only a loss of physical *information* -a left foot - when compared to Dion.

In addition, on premise that if Theon & Dion were really two men, Dion's left foot amputation would still not make Theon narratively identical to Dion because both Theon and Dion on premise if existing, must be living in different spaces in space-time continuity. This distinction in the *spatial-temporal information* is what makes Theon different from Dion. We can encounter the same argument without Theon but rather only Dion and his left foot.

What is Dion's identity over time? We cannot answer this question without questioning the differences amongst Dion yesterday with a left foot, Dion today with an amputated left foot, and Dion tomorrow with still an amputated left foot. Dion yesterday is completely a left foot more than Dion today; thus, they cannot be the same; but how about Dion today and Dion tomorrow? Are they the same? Or rephrased in another way, is today Dion's identity conserved for tomorrow?

Upon discussion laws of conservation, Noether's theorem has mathematically proved that conservation of any quantity in any system is the direct consequence of symmetry in that system (Hanc, Tuleja, and Hancova). Noether's theorem is exceptionally helpful for scientists to know whether a quantity is conserved by examining the symmetry in question. In identity crisis, if we want to know whether today Dion is the same with tomorrow Dion, both with a lost left foot, we must examine via the axis of symmetry in the space-time continuum. If we say the narrative identity of objects is composed of information, we should evaluate the information associated with today Dion versus information associated with tomorrow Dion to see if there is any symmetry between the two in the space-time continuum. This evaluation asks if there exists any moment that not only Dion today is the same with Dion tomorrow but also *everything else* is the same, meaning every bit of information must represent the same, suggesting that the positions of micro objects

such as atoms and positions of macro objects such as dogs, cats, cars, etc. between today and tomorrow must match. If there exist such moments, it means there is a symmetry in space-time continuum and the identity between Dion today and tomorrow is conserved. Indeed, if somebody asks me what the difference is between today and tomorrow, I will say that tomorrow means something *changes* versus today. If tomorrow has nothing *changed* compared to now, I wonder whether it should be called tomorrow, for which no changes only indicate no time flow. In these kinds of scenarios, it is almost *impossible* for Dion today to be narratively identical to Dion tomorrow because within a small scale of time like today and tomorrow, the chance of everything to coincide is essentially *near-zero*. As a result, Dion today and Dion tomorrow is almost *guaranteed* to be *narratively different*. Now, should one scale the period up and ask the question again, such that is Dion today and Dion 80 years later the same (given that Dion is still alive), there will be more coincidence between information of Dion today and Dion 80 years later, but it is hopelessly low, *near-zero* still. To add more flavor in the heat of discussion, one may ask how about *infinite* time? **Had Dion been immortal so that Dion could question his identity after *infinite* amount of time, every combination in the universe would be able to replicate at least once and Dion's identity would conserve and converge thanks to the *symmetrical* nature of the mentioned scale.** In the other words, if there was an infinite amount of time for a person to live in, his or her identity would be identical to one that many people know quite well: God. An immortal God will necessarily be *omnipresent*, as he has the identities of anything at any moment. In information theory of identity, immortality necessarily and sufficiently equates God to *anything*, reflecting that mortality equates identity to *uniqueness* and thus entails that no two objects are narratively identical.

My observation regarding to this matter is not unique, as I share my view with Wolfgang Pauli, who proposed a famous and brilliant scientific principle in quantum physics called “Pauli exclusion principle” (Pauli). This principle states that in an atom or molecule, no two electrons can occupy the same quantum numbers. These four quantum numbers are n , l , m_l , and m_s . Each number of this set represents a kind of *information* about the electron. Pauli’s principle suggests that *no two electrons can have identical quantum state*, suggesting that no two electrons can carry the same informational state.

In articulating further this informational reading of quantum uniqueness, it is mandatory to distinguish between two broad categories of information: natural information and artificial information. Natural information refers to informational features of the world that appear independent of human construction, which persist whether any mind apprehends them. Examples include the spatial arrangement of atoms in a crystal lattice, the motion of planets, the genetic sequence of an organism, or the time elapsed during a physical process. These informational structures *seem* to belong to the world itself, operating with or without human awareness. By contrast, artificial information encompasses the forms of information humans deliberately construct to organize, transmit, and interpret aspects of reality. This category includes linguistic systems, grammatical rules, mathematical formalisms, scientific classifications, philosophical taxonomies, and the content of books or databases. Yet this distinction between “natural” and “artificial” raises a deeper philosophical tension, one that George Berkeley expressed in *A Treatise Concerning the Principles of Human Knowledge*. Berkeley famously denies that the so-called ‘objective world’ can exist independently of perceiving minds. He argues that what we call the material world consists solely in ideas, *esse est percipi*, meaning “to be is to be perceived.” As he writes, **“the existence of an idea consists in being perceived” (§3 Berkeley), and therefore the**

very notion of material objects existing unperceived is unintelligible. Furthermore, Berkeley maintains that all the qualities we attribute to external objects: shape, motion, solidity, are inseparable from perception, insisting that “those things which are called sensible qualities... cannot exist otherwise than in a mind perceiving [them]” (§14 Berkeley). From this standpoint, what I have called “natural information” begins to look suspiciously like a projection of stable patterns across shared experiences, rather than a reservoir of mind-independent *truths*. Even the claim that the spatial arrangement of atoms persists without observation would, for Berkeley, amount to a category mistake, since “**the absolute existence of unthinking things are perfectly unintelligible, and involve all the absurdity of abstraction**” (§24). In short, Berkeley pushes us to confront how much confidence we can place in informational structures we take to be “objective,” given that all such structures are accessed, and perhaps constituted, through the interpretive activity of minded creatures. From Berkeley’s point of view, it becomes exceedingly difficult to secure any definite answer concerning the *metaphysical identity* of objects, which is the identity they possess independently of any perceiving subject. If identity is composed of informational content, and if all informational content is delivered through perception, then the very boundary between what I have called “natural” and “artificial” information dissolve rapidly. The project of isolating a mind-independent informational core that could ground the intrinsic identity of objects falters once we acknowledge Berkeley’s challenge. Even the most seemingly natural informational features are available to us only through perception and therefore cannot be separated from the interpretive frameworks our minds impose.

It seems that resolving this tension requires choosing between two unappealing options. The first is to reject Berkeley’s argument concerning mind-dependent existence and to defend the claim that material objects persist as intelligible entities even when unperceived. The second is to

accept the existence of material objects that continue to exist unperceived, yet whose existence in that state is wholly unintelligible to us. Many philosophers, naming David Hume, Thomas Reid, Immanuel Kant, etc., have taken the first path and have advanced extensive criticisms of Berkeley's idealism, and that debate remains ongoing. I seek not to add to those arguments here. Instead, I call my luck in consideration of the second possibility and examine its implications. If we accept that objects exist independently of the mind, then we must also embrace that they exist in a condition that is **entangled, unable to comprehend, and unintelligible** from the standpoint of human cognition. In this condition, the external world is not yet shaped by perception, concept, or interpretation and thus cannot be understood as carrying *information* in any meaningful sense. Rather, such objects exist as undifferentiated blocks of *data*, awaiting the interpretive act through which minds transform data into information. If I were to be logical, it is this block of data is the answer we have been seeking for: ***The Metaphysical Identity of Everything.***

Adapting this viewpoint entails several peculiar consequences for our understanding of the world. One of the most striking is that it becomes impossible to comprehend, or even properly identify, the origin of the universe (or multiverse). Any attempt to do so would contradict the premise that the fundamental source of all identities, the metaphysical condition in which objects exist prior to perception, is unintelligible to the human mind. In this primordial condition, reality exists only as unstructured data, and the moment human cognition attempts to grasp it, this data is inevitably transformed into directed and meaningful information. The result of this interaction is one narrative identity amongst many, or I would say "One Form Over Many" (Plato Rep. 596a). If this analysis is sound, then what I have been calling the metaphysical identity of objects can be viewed as the very Form, or even the soul, that Plato sought in his inquiries. For Plato, a Form is the fundamental principle that confers unity and identity upon a diversity of instances. It is the

stable reality beneath the shifting appearances of the sensible world, or I shall call *the foundation of The World*.

If we continue to explore the implications of this metaphysical framework, we find ourselves confronting results that would once have seemed unbelievable. One such consequence is that certain psychic or telepathic phenomena, long regarded as curiosities or superstitions, become intelligible. If all minds draw from, and interact with, a shared layer of pre-perceptual data, then it is not entirely incoherent to suppose that **some individuals might experience overlapping informational states**. Among the most frequently reported of such phenomena are the striking co-experiences documented between twins, especially identical twins. The psychical literature contains numerous accounts in which twins appear to sense, feel, or envision events occurring to the other, even across great distance, in ways that defy ordinary causal explanation.

One particularly vivid case appears in Frederic W. H. Myers's *Human Personality and Its Survival of Bodily Death*. Myers recounts the testimony of Mrs. Storie, whose deceased brother was also her twin. On the night of his fatal accident, she experienced an intensely detailed and emotionally charged dream-vision in which she saw him injured upon a railway line, accompanied by a profound sense of dread. She later learned that her brother had died in precisely the circumstances she had envisioned. Myers emphasizes that her notes were written immediately upon learning of the death, lending evidential weight to the account, and he explicitly notes that her brother being a twin "adds interest to the case, since one clue to the causes directing and determining telepathic communications lies in the exceptional frequency between twins" (p. 144).

A second case, also recorded by Myers, concerns James Carroll, a butler personally known to Edmund Gurney. Carroll reported a sudden and vivid apparition of his twin brother appearing before him late at night, an image so striking that it halted him in mid-action and left him

momentarily stunned. Only moments later, his real brother arrived at the house unexpectedly, tapping on the window and calling to him. Carroll further testified that he had previously experienced powerful sensations of exhaustion and sadness precisely at the onset of his distant twin's fatal illness, impressions he could not rationally account for. Myers notes that the link of twinship 'seems markedly to facilitate this kind of communication,' drawing on Carroll's experience as a compelling example of twin-related psychical concordance (p. 272).

These cases are impressive to me, for they exhibit a level of experiential coordination that seems almost impossible among ordinary individuals, yet appears with surprising regularity among monozygotic twins. As I consider these accounts, I notice that explanations for such phenomena tend to fall into familiar arguments. Some interpret them as signs of divine intervention, suggesting that events unfold with an intended purpose. Others, especially atheists or naturalists, argue that if these experiences are not the result of supernatural forces, they must arise from coincidence, perhaps embellished by memory, narrative drift, or selective reporting. In this line of reasoning, "selective reporting" plays a crucial role. Human beings naturally attend to, remember, and retell events that are unusual, emotionally charged, or narratively satisfying, while ignoring the vast majority of mundane or contradictory cases. As a result, stories of twins experiencing uncanny coincidences are far more likely to be preserved, repeated, and embellished than the countless instances in which twins do not sense anything unusual at all. A day in which nothing remarkable happens rarely enters the record, but a rare and emotionally striking moment does. Over time, this asymmetry in what is remembered and communicated generates an illusion that the phenomenon is more common than it truly is. What survives in cultural memory is therefore not a representative sample of all twin experiences, but only the *extraordinary* ones, which biases our perception and invites explanations that reach beyond the ordinary. Nevertheless, I believe it comes as no surprise

that the extraordinary, the exceptions, or the minority helps to reveal more secrets than the ordinary could ever have.

In explaining these marvelous stories, I believe they are in part the result of an overlap within the narrative identities of the twins involved. In a scientific sense, the act of seeing is not magical but simply a physical interaction between photons and the photoreceptors of the eye, which then generates neural activity that we interpret as visual perception. Yet in circumstances where synapses in the brain interact without any corresponding stimulation of the photoreceptors, the brain can produce hallucinatory images that appear vividly real to the subject. These hallucinations are normally regarded as unreal because they are not corroborated by the shared physical environment in which observers ideally perceive the same objects. But as I said in another section of this paper, **information does not construct reality, it is reality**; and believe it or not, the interactions between synapses inside the brain is also a kind of information, it is **equally real** as information obtained from the photoreceptors. You can see where I am going to, for that I am saying twin telepathic might be possible, and those visions are the direct consequence of internal movements of the brains. This hypothesis is further supported by the fact that telepathy happens much more frequently between monozygotic twins (Myers), where they generally have the same pools of information, naming genetically and most of the time environmentally too. When chance allows, any random internal movements between these twins might coincide and encode for the same "vision," explaining for the special bonds between identical twins. Frankly speaking, we should only expect to see these kinds of interactions between close friends, as they share somewhat similar "narrative identity," whose random deviations tend to be similar perceptive realities, or as termed in this paper: similar narrative identities. As a result, these people have the same "vision" that we call telepathy.

Intrigued by documented cases like ones mentioned by Myers, scientific communities have invested intense efforts in discovering telepathy and most of them conclude that telepathy is nothing besides coincidence. In “An Experimental Investigation of Telepathic Phenomena in Twins,” Kubis and Rouke designed multiple experimental procedures to probe whether twins possess a special capacity for direct mental communication. Their central method involved having each pair of twins attempt to identify symbols on Zener cards while separated from both the agent and from each other. Each twin’s guesses were compared both to the actual card sequence and to the other twin’s responses to determine whether their choices corresponded at a rate higher than chance. A total of six twin pairs were tested, with each pair completing fifteen hundred trials under controlled conditions.

Despite the scope of the testing and the occasional appearance of small positive correspondences, none of the twin pairs demonstrated results large enough to surpass the required statistical critical value. Most deviation scores, whether above or below chance, remained well within the boundaries expected from random variation. Because meaningful evidence of telepathy would require correspondence values significantly exceeding chance probability, the observed fluctuations were interpreted as coincidence rather than genuine communication. As a result, Kubis and Rouke concluded that the data provided no scientific support for the existence of twin telepathy.

Although the twin telepathy sessions involving simultaneous guessing yielded results consistent with chance, the individual-agent tests assessing what researchers call Extra Sensory Perception, or ESP, present a more complicated picture. ESP refers to the purported ability to acquire information without reliance on the recognized sensory modalities, and includes phenomena such as clairvoyance, telepathy, and precognition. In the twin study conducted by Kubis and Rouke, several of the individual tests revealed performance that modestly exceeded

chance expectation, with a few experimental runs producing odds against chance as high as twenty thousand to one (p.43 and 44). These outlier results indicate that under certain conditions a single percipient, whether one member of a twin pair or possibly both, may display patterns of extra-sensory perception not easily dismissed. It is important to note that these one-sided anomalies resemble the cases documented by Myers, where the communication did not proceed in a reciprocal fashion but manifested one-sided projection from one twin to another. Within the interpretive framework developed earlier in this paper, such one-sided episodes can be understood without invoking supernatural causation. If narrative identity emerges from the continuous fluctuations of informational activity in the brain, then rare alignments of these fluctuations may generate internally produced visions or impressions that appear paranormal. The chance concurrence of these internal states between two genetically and often developmentally similar individuals, such as monozygotic twins, appear at much higher rates than between non-similar individuals, producing the illusion of telepathic transmission while still remaining compatible with a naturalistic explanation grounded in the informational structure of the mind.

Despite the striking odds reported in the exceptional trials, Rouke is careful to emphasize that these results do not indicate a stable or reliable capacity for ESP in any of the participants. As he makes clear in his analysis, the unusually high scores emerged only under particular conditions, such as reduced fatigue, diminished emotional strain, or moments of heightened relaxation (p.44). Outside these narrow circumstances, performance returned to levels fully compatible with chance. Yet Rouke's caution neither undermines nor bolsters the hypothesis I have been developing. His conclusion merely states that whatever produced these anomalous scores is inconsistent, but it does not address the deeper question of what underlying mechanism could generate such irregularities in the first place. From my perspective, these episodic successes remain entirely compatible with

the hypothesis that telepathic-like episodes and one-sided ESP manifestations arise from random fluctuations in neural activity. If narrative identity and experiential content are shaped by dynamic informational processes within the brain, then it would not be unexpected that, on rare occasions and under certain conditions, these processes align in unusual ways, producing brief moments that appear extraordinary. Kubin & Rouke's findings do not contradict this possibility, for what they found is an increase of likelihood that chance in the Zener cards alone would not achieve with an odds against chance of twenty thousand to one. I agree that monozygotic twins are very *similar* but real-life scenarios might be so complicated that even genetical and environmental similarity are just ones of many informational sources, rendering coincidental vision a low but non-zero probability. To further illustrate the idea of identity convergence, in the few next paragraphs, we will talk about something even closer than twins: different versions of oneself in time and dissect *déjà vu* phenomenon, not as a *malfunction* of the brain but rather a *paranormal* activity where identities across time converge.

In *The Déjà Vu Experience*, Alan Brown presents one of the major contemporary explanations for *déjà vu* as the dual-processing model, which proposes that the phenomenon arises when two cognitive pathways that ordinarily operate in synchrony momentarily fall out of alignment. As Brown describes it, "the central idea of dual-processing theories is that *déjà vu* is the result of two otherwise parallel and synchronized cognitive processes becoming momentarily out of synch" (p. 19). This account fits well with the observation that *déjà vu* carries an immediate sense of familiarity that lacks a corresponding act of recollection. Yet, as Brown notes, the model faces substantial limitations. Although influential, "the dual-processing theories are...difficult to test" (p. 33) because the very processes they posit are internal, abstract, and resistant to direct measurement. Consequently, researchers have found it challenging to design experiments that

capture the proposed desynchronization or reproduce the experience in controlled conditions. The theory therefore offers a partial explanation, accounting for the feeling of familiarity but falling short of addressing the emotional force and structured depth that often accompany déjà vu, whose nature is richly layered with emotions, backgrounds, visions, etc. and exceeds what a brief desynchronization between parallel cognitive streams could plausibly generate, leaving the deeper experiential texture of déjà vu largely unexplained within the dual-processing framework.

A second major explanatory framework arises from clinical observations in epilepsy, particularly temporal-lobe epilepsy. Brown notes that many epileptic patients report experiencing déjà vu immediately before the onset of a seizure, a pattern that has led some researchers to regard the phenomenon as a by-product of temporal-lobe malfunction rather than as a normal cognitive event. As he explains, “déjà vu is one of the more frequent experiential correlates of temporal-lobe seizures” (Brown, p. 24), which suggests a possible neurological basis. Yet this account faces an immediate difficulty: it does not generalize to the vast majority of healthy individuals who experience déjà vu without exhibiting any measurable neurological pathology. In response to this line of reasoning, I would argue that the epileptic form of déjà vu is best understood not as the paradigm case but as a pathological imitation of it. Since the temporal lobe plays a central role in processing temporal order and continuity, its disruption in epilepsy would naturally produce confusion about the sequence of events. This confusion can generate what I call a “false positive” déjà vu experience, wherein the individual misinterprets distorted temporal processing as familiarity. In other words, epileptic déjà vu arises from a malfunction in the temporal-lobe machinery that ordinarily structures our sense of temporal flow and therefore should not be taken as representative of the genuine déjà vu experienced by healthy individuals.

Beyond these two dominant models, Brown catalogs a wide variety of additional scientific theories attempting to account for déjà vu, ranging from memory-familiarity mismatches to holographic memory models, divided attention accounts, attentional misfires, and even cryptomnesia-based explanations (Brown, pp. 17–35). The sheer diversity of these proposals reveals a central problem: there is no universal or even widely agreed-upon scientific explanation for the phenomenon. Each theory accounts for a limited subset of cases, but none succeeds in offering a comprehensive account that captures the full phenomenology of déjà vu across healthy populations. This lack of theoretical convergence has led me to suspect that déjà vu, at least in its deepest form, may not be reducible to a purely neurological or purely psychological glitch. Instead, within the metaphysical framework developed in this paper, déjà vu appears more plausibly as a momentary coincidence of identities, where the informational configuration of the present self aligns with that of a future self. When the future event eventually occurs, the individual “**hallucinates**” this convergence for a prior memory and thus experiences the powerful illusion of having already lived the moment before. In this sense, déjà vu becomes not an error in neural timing but a brief collapse of temporal narrative identity across time, producing an uncanny sense of standing simultaneously in the present and in a future that has not yet unfolded.

My proposal is not without supportive evidence, for aspects of my account align surprisingly well with contemporary cognitive science. One particularly relevant body of work is Daniel Schacter’s constructive memory theory, which holds that memory is not a literal retrieval of stored data but a constructive act in which the mind dynamically rebuilds past events from fragmented informational traces. Schacter and colleagues propose that our episodic memory system enables the flexible retrieval and recombination of past experience for the purpose of simulating possible future events (Schacter et al., 2008), a view that fits neatly with my hypothesis

that *déjà vu* emerges when the construction process momentarily aligns present informational states with those that properly belong to a future moment. If memory is inherently reconstructive rather than reproductive, then the boundary between past and future informational patterns becomes more permeable than traditional models allow.

This idea is reinforced by the work of Daniel Schacter, Donna Rose Addis, and Randy Buckner on what they call the “prospective brain.” Their research demonstrates that the cognitive and neural systems responsible for remembering the past are the very same systems recruited when we imagine the future. As they explain, the brain relies on “a constructive memory system that flexibly recombines details from past experiences” in order to simulate events that have not yet occurred (Nature Reviews Neuroscience 8, 2007, p. 659). In this framework, both memory and future thought arise from a shared constructive architecture rather than from two fundamentally different mechanisms. The mind does not simply retrieve stored records of the past; it actively reassembles fragments of experience into coherent mental scenes, whether oriented backward or forward in time. If so, *déjà vu* can be interpreted as a moment in which these constructive processes intersect, causing the present perceptual episode to be misclassified as one of the brain’s internally generated future simulations. What feels like “remembering” may instead be the accidental recognition of a future-oriented construction produced earlier by the same neural system that handles episodic memory.

Neuroimaging evidence provides even stronger support. Functional MRI studies repeatedly show that remembering the past and envisioning the future activate nearly identical neural networks, particularly within the medial temporal lobe, hippocampus, and prefrontal cortex. Schacter, Addis, and Buckner note that “the neural overlap between remembering the past and imagining the future is striking,” emphasizing that the same core regions of the default network

support both processes (Nature Reviews Neuroscience 8, 2007, p. 659). Their research indicates that episodic memory and episodic future simulation draw on a unified system dedicated to temporal self-projection rather than separate mechanisms. The fact that the brain relies on this common circuitry to retrieve old events and to construct imagined ones implies that the sense of familiarity characteristic of déjà vu may arise when the brain accidentally treats a forward-directed simulation as if it were a backward-directed memory. When the informational configuration of the present moment resembles a neural representation, the mind has already assembled while imagining a possible future, the experience of déjà vu becomes a natural byproduct of this **neural symmetry** (Schacter, Addis, and Buckner 660).

Taken together, these findings lend me plausibility to the idea that déjà vu reflects a moment of identity convergence between temporal versions of the self. Cognitive science already acknowledges that the mind constructs both past and future within the same neural architecture, so my suggestion that déjà vu involves a brief overlap between present and future informational states is not as metaphysically extravagant as it first appears. Rather, it reframes déjà vu as a rare but meaningful event in which the constructive machinery of the mind momentarily collapses the distinction between *what has happened* and *what is yet to happen*. With these considerations in place, I am now led to question whether the deeper source of déjà vu lies not in memory at all, but in the nature of time itself, for I increasingly suspect that time is an illusion whose “future” and “past” exist only as constructions produced by the mind.

Talk about the illusions of time: the past, present, and future.

Talk about dreams.

“The telepathy activities between twins: seeing / feeling the *same* thing, or time traveling,
or dopperganger.

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